Granular Disintegration:

Granular disintegration takes place in crystalline rocks where the grains of the rocks become loose and fall out. This is due to the action **of temperature**.



Block Disintegration:

Repeated expansion and contraction of rocks during day and night respectively causes stress on the joints of the rocks which results in block disintegration



Hots

Is weathering a pre-requisite in the formation of soil?

Facts

The disintegrated rock materials, in due course of time, are weathered further, to form soil. Soil is a mixture of disintegrated rock material and decayed organic matter called humus.

Chemical Weathering

Disintegration and decomposition of rocks due to chemical reactions is called Chemical Weathering. This is predominantly high in the hot and humid regions such as the equatorial, tropical and sub tropical zones. Chemical weathering takes place through the processes of **oxidation, carbonation, solution,** and **hydration.** The agents of Chemical weathering are Oxygen, Carbon-dioxide, Hydrogen and water.



Oxidation

Oxygen in the atmosphere reacts with the iron found in rocks, thus leads to the formation of iron oxide.This process similar to the rusting of iron, pressure of air and water is known as oxidation, which results in the weakening of rocks.

Carbonation

Carbonation is the mixing of water with the atmospheric carbon-dioxide, forming carbonic acid. Carbonation is important in the formation of caves, in limestone region. When the carbonic acid reacts with the carbonate rocks, the rocks get disintegrated.

Solution

The dissolution of rock substances in water result in the loosening of the rock particles. This inturn breaks down the rocks.

Hydration

Absorption of water into the mineral structure, certain chemicals in the rock enlarge in size in humid conditions. These minerals found in the rock swell and this results in the development of cracks and the

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